Abstract Algebra

Based on personal study and various lecture notes

Yi Wang

NOTES ON MATHEMATICAL STUDIES
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Preface

Hello, I am a preface.

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Contents

CHAPTER 1

Groups

1.1 Cyclic Groups

Definition 1.1.1. Hello, I am a definition.

Lemma 1.1.1. *Hello, I am a lemma.*

Theorem 1.1.1. *Every cyclic group is abelian.*

By Theorem 1.1.1, any two elements of a cyclic group commute. For these basic properties of cyclic groups and their subgroups, see Dummit and Foote [DF04].

Detailed proof (p. 1)

Corollary 1.1.1. *Hello, I am a corollary.*

Proposition 1.1.1. *Hello, I am a proposition.*

Example 1.1.1. Hello, I am an example. ◇

Remark 1.1.1. Hello, I am a remark. ◇

Problem 1.1.1. Hello, I am a problem.

Proof. Hello, I am a proof. □

By Exercise 1.1.1, every cyclic group is abelian.

Proof of Theorem 1.1.1. Let $G = \langle g \rangle$, and take arbitrary $x, y \in G$. Then there exist $m, n \in \mathbb{Z}$ such that $x = g^m$ and $y = g^n$. Hence

$$xy = g^{m+n} = g^{n+m} = yx.$$

Therefore G is abelian. □

Return to original

Exercise

1.1.1. Prove that every cyclic group is abelian.

1.1.2. Prove that every subgroup of a cyclic group is cyclic.

1.1.3. Let $G = \langle g \rangle$. Prove that $G = \langle g^{-1} \rangle$.

1.2 Symmetric Groups

Chapter 2

Rings

2.1 Subrings

Appendix A

Reference Answers

1.1 Cyclic Groups

1.1.1. Every cyclic group is abelian.

1.1.2. Every subgroup of a cyclic group is cyclic.

1.1.3. If $G = \langle g \rangle$, then $G = \langle g^{-1} \rangle$.

Appendix B

Set Theory

B.1 Basic Set Theory

Hello, I am an appendix.

Appendix C

References

[DF04] D. S. Dummit and R. M. Foote. *Abstract Algebra*. 3rd ed. Wiley, 2004.